Dr. Yankuba B. Manga (楊庫巴), PhD

助理教授 Assistant Professor | 國立臺北大學 National Taipei University

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Professional Profile

- Dr. Manga serves as an Assistant Professor at National Taipei University and the Director of the Smart, Health, AI & Leadership (SHAIL) Lab. His research centers on artificial intelligence applications in healthcare, specifically in clinical decision support, medical imaging analysis, and precision medicine.
- His expertise encompasses machine learning, deep learning, and explainable artificial intelligence (XAI), with a focus on integrating multimodal clinical data, such as medical imaging and electronic health records, to enhance disease risk prediction and survival analysis. His work seeks to improve the transparency, reliability, and clinical utility of AI systems in healthcare settings.
- He has substantial experience in data governance, research ethics, and the responsible implementation of AI technologies in medical and public health domains. Through interdisciplinary collaboration, his research advances evidence-based, trustworthy, and scalable AI solutions for contemporary healthcare systems.

Academic & Professional Experience | Taiwan

❑ 國立臺北大學 | National Taipei University — 2023.08 – Current

– Master Program in Smart Healthcare Management

- **Position** – Assistant Professor
  - Research and Teaching; Supervising BSc/MSc Students

❑ 臺北醫學大學 | 醫學工程學院 Taipei Medical University — 2019.01 – 2021.12

– School of Biomedical Engineering | Biomedical Devices - Bone & Joint Research Lab

- **Position** – Postdoctoral Researcher
  - Led NHRI-Funded Hydrogel Drug-delivery Project; Supervised MSc/PhD Students

Education | Taiwan

❑ 臺北醫學大學 | 醫學工程學院 Taipei Medical University — 2010.09 – 2019.01

- **Degree** – Doctor of Philosophy (PhD) [Biotechnology] — 2019.01
  - **Research** – "High-Throughput Screening Through Poly-SiNW FETs for Ultra-sensitivity and Selectivity, Label-Free and Real-Time Biosensing"
- **Degree** – Master of Science (MSc) [Biotechnology] — 2013.01
  - **Research** – "Research on Pathogenic Bacteria Decontamination by High Electric Field - Nonthermal Plasma for Biomedical Applications"

❑ 國立臺北科技大學 National Taipei University of Technology 2007.09 – 2010.06

- **Degree** – Bachelor of Science (BSc) [Electronic Engineering – Computational Systems]
  - **Project** – Computational Systems

❑ 銘傳大學 Ming Chuan University — 2005.09 – 2007.06

- **Degree** – Undergraduate Program — 2006.09 – 2007.06
- **Certificate** – Mandarin and Chinese Culture Studies — 2005.09 – 2006.08

Publications

❑ Selected Peer-reviewed Articles

- Wu, C.E., & **Manga, Y.B.*** (2025). Impact of wearable-assisted walking on sarcopenia and body composition in older adults. BMC Geriatrics, 25(1), 466. https://doi.org/10.1186/s12877-025-06142-x
- Asri, Y., … , **Manga, Y.B.*** (2025). Prevalence and associated factors of loneliness among older adults in Indonesia: Insights from the Indonesian Family Life Survey (IFLS-5). Jurnal Ners, 20(1), 9–16. https://doi.org/10.20473/jn.v20i1.59927
- Asri, Y., … , **Manga, Y.B.*** (2024). Sleep quality and cognitive function on self-rated health status among the elderly: Findings from the Indonesian family life survey (IFLS-5). Narra J, 4(3), e1103. https://doi.org/10.52225/narra.v4i3.1103
- Mai, H.X., … , **Manga, Y.B.*** (2024). MT13 Comprehensive Assessment for Breast Cancer Diagnosis Employing MATLAB Pre-Trained Models with Machine and Deep Learning. Value in Health, 27(6), S283. https://doi.org/10.1016/j.jval.2024.03.1551
- Asri, Y., … , **Manga, Y.B.*** (2024). Depression Among Islamic Boarding Schools Students During the COVID-19 Pandemic in East Java, Indonesia. Kesmas: Jurnal Kesehatan Masyarakat Nasional, 19, 51. https://doi.org/10.21109/kesmas.v19i1.7437
- Putri, A.D., … , **Manga, Y.B.** (2024). Whole-Cell Electrochemical Aptasensors for Cancer Diagnosis. Adv. Sensor Research, 3(5), 2300151. https://doi.org/10.1002/adsr.202300151
- Full List Available:
  - https://orcid.org/0000-0002-8750-3721
  - https://www.scopus.com/authid/detail.uri?authorId=57195972825

❑ Conferences | Guest Lectures | Speaker

- **Manga, Y.B.** (2025). AI Applications in Healthcare, Medicine, and Public Health, Surabaya, 28th October, 2025 – Online – Guest Lecturer
- **Manga, Y.B.** (2025). ICORE 2025 International Conference on Computing, Research, and Engineering, Institute of Technology Al-Mahrisiyyah Lirboyo, Kediri, Indonesia – Guest Speaker
- Elmi, K.H., Wu, C.-E., & **Manga, Y.B.** (2025). Reversing sarcopenia with wearable-assisted walking in older adults. UHIMA & TLCMA Symp., Kaohsiung.
- Bianchini, G., Adeel, M., & **Manga, Y.B.** (2025). AI in emergency care decision-making: A systematic review. UHIMA & TLCMA Symp., Kaohsiung.
- Mai Xuan, H., Duong-Cao, T.T., & **Manga, Y.B.** (2024). Breast cancer diagnosis via MATLAB pre-trained models. ISQua 2024, Istanbul.
- Pawarti, H., Adeel, M., & **Manga, Y.B.** (2024). Digital transformation in cosmeceutical skincare: Gap mapping. FAPA 2024, Seoul.
- Mai, H., Cao, T., & **Manga, Y.B.** (2024). MATLAB-based breast cancer assessment. Value in Health, 27(6), S283, USA.

Scholarly | Achievements

❑ Teaching | Mentorship

- **Graduate Courses** – Smart Healthcare Management, AI Health Systems, Digital Transformation.
- **Workshops** – Scientific Writing, EndNote/Mendeley, SPSS/GraphPad, AI Ethics, Systematic Reviews.
- **Mentored | Supervised –** 30+ MSc/PhD Students.

❑ Leadership | Advisory | Interests

- **Founder –** SHAIL Lab – Leads Research on AI, Data-driven Healthcare, and Leadership Development.
- **Editorial Board –** Computer Methods & Programs in Biomedicine Update.
- **Reviewer –** Cureus, IJ Nanomedicine, ACS AMI, BMC, ELSEVIER, etc.
- **Associate Member** – Cheeky Scientist Association–World Only PhDs
- **Consultancy –** Scientific & Technical Writing, Statistical & Analytical, etc.
- **Co-Founder –** GamSTEM, African Student Organization, TMU Muslim Student Association
- **Advisor –** GamTai Association; Dormitory & Welfare Officer for International Students (TMU).
- **Sports & Well-being Advocate –** Fitness, Former Soccer Assistant Captain/Manager.

Scholarly Achievements

- Distinguished Talent in Smart Healthcare Management, NTPU (2023-2025)
- Excellence in Teaching and Mentor, NTPU (2024)
- TMU Postdoctoral Research-NHRI Project Award
- TMU International Scholarships – MSc/PhD
- MOFA Fellowship/Scholarship.

Languages | Technical Expertise | Skills

- **English** [Native] **Mandarin | Chinese** [Professional Level] **French** [Basic]

Technical Expertise

- **Data Analysis & Computational Methods**
  - Programming and Analytics – MATLAB, SAS, Python, R (Anaconda)
  - Statistical Analysis – SPSS, SmartPLS 4
  - Data Visualization and Reporting – Power BI
  - Research Productivity Tools – Microsoft 365, Academic Reference Management Software
- **Scientific & Clinical Research Platforms**
  - Experimental Design – *in vitro*, *ex vivo*, and *in vivo* Study Design and Execution
  - Translational Biomedical Research – Integration of Lab Findings into Clinical and Applied Research
  - Medical Device Evaluation – Assessment and Validation of Diagnostic and Therapeutic Devices
  - Sports and Rehabilitation Medicine Applications – Biomechanical and Physiological Analysis
- **Medical Devices & Bioengineering**
  - Electrochemical Biosensors – Design, Surface Modification, Functionalization
  - Microfluidic Systems – Lab-on-chip Platforms Development and Fluidic System Optimization
  - Implantable and Minimally Invasive Devices – Design, Simulation, and Rapid Prototyping
  - Drug Delivery and Modification Systems – Controlled Release and Formulation Optimization
- **Materials Science & Multiphysics Modeling**
  - Functional Nanomaterials – Synthesis, Characterization, and Biomedical Applications
  - Multiphysics Simulation – COMSOL-based Modeling for Coupled Physical Phenomena
  - Tissue Engineering – Scaffold Design, Optimization, and Performance Evaluation
- **Imaging, Spectroscopy & Quantitative Analysis**
  - Advanced Imaging Techniques – Fluorescence, and Confocal Microscopy
  - Surface and Structural Characterization – SEM, TEM, AFM
  - Spectroscopic Techniques – FTIR, and Raman Spectroscopy
  - Quantitative Image Analysis – Feature Extraction, Segmentation, and Scientific Visualization

Professional & Research Skills

- Scientific Leadership and Project Coordination in Interdisciplinary Environments
- Research Design, Execution, and Methodological Troubleshooting
- Team Leadership, Collaboration, and Personnel Management
- High Adaptability and Rapid Learning in Emerging Technologies
- **Scientific Communication** – Academic Writing, Presentations, Stakeholder Engagement

NTPU Courses Offering

- Computer Programming 程式設計
- Data Mining 資料探勘
- Machine Learning 機器學習
- AI and Data Sciences 人工智慧與資料科學
- Research Methodology 研究方法
- Academic Writing 學術寫作
- Biotech & Medical Industry Strategy & Decision Analysis 生技醫療產業策略與決策分析
- Calculus 微積分 – Undergraduate Level
- AI Application in Health Economics and Financial Management 健康經濟與財務管理之人工智慧應用 – Undergraduate Level

End! 202512